COURSE 1 - IT APPLICATION OVERVIEW AND FEATURES

1.1 Objective

- Introduction to Application Understanding
- Different types of Applications with Examples
- Basic Application Architecture

1.2 Course Content

1.2.1 Introduction to Application Understanding

If you manage your work without any mess with the help of alarms in a smart phone

Figure 1

Or if you are reading any posts with your tablet,

Figure 2

You are using Applications/App.

So what exactly does an Application mean?

An application is a type of software that allows you to perform specific tasks which otherwise is very difficult to perform manually.

Before we go any further think about few computer applications you are familiar with

Sticky Notes

You can use Sticky Notes to write a to-do list, jot down a phone number, or to do anything else that you would you use a pad of paper for.

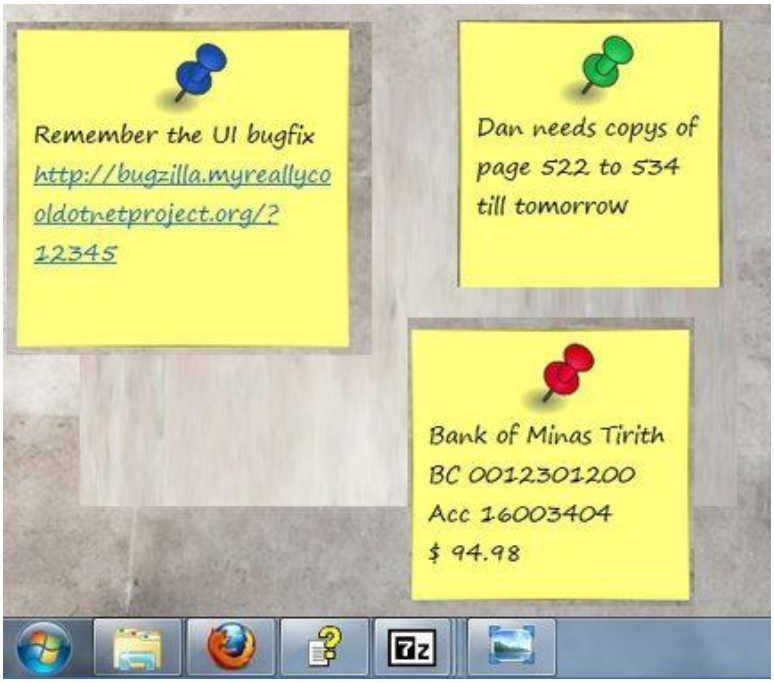


Figure 3

Windows Notepad

Notepad is a basic text editor that you can use to create simple documents.

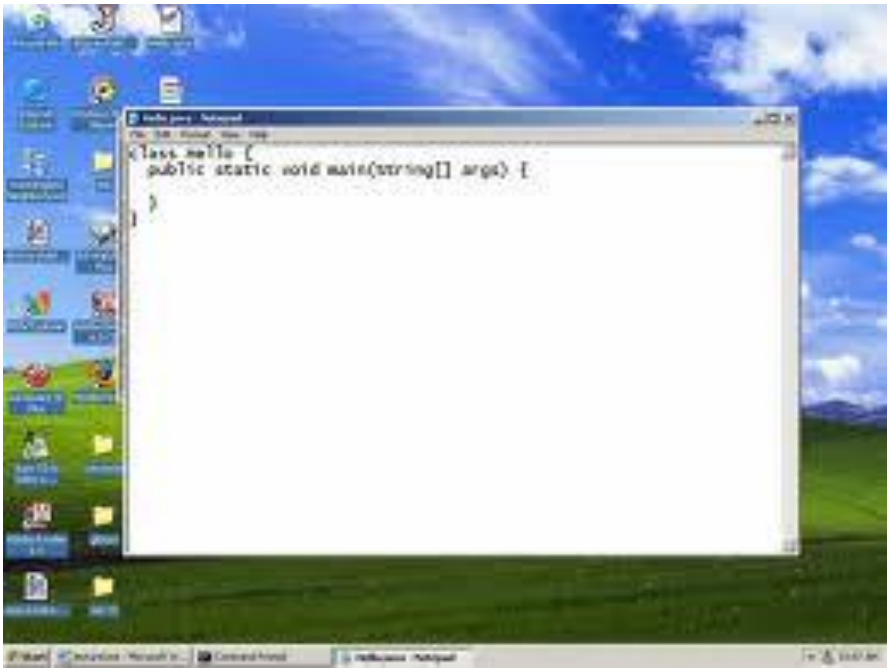


Figure 4

Whatsapp

WhatsApp Messenger is a proprietary, cross-platform instant messaging subscription service for smart phones.

Figure 5

Facebook Application

A social utility that connects people with friends and others who work, study and live around them

Figure 6

IRCTC Portal

Online Passenger Reservation System provides booking facility of Railway tickets online

Figure 7

By giving a closer look to these applications you will find that few of the applications are specially designed for Desktop/Personal computer while few others only work on mobile devices. Similarly you will find that in order to run certain other applications you need a web browser like Internet Explorer, Chrome etc.

1.2.2 Different types of Applications with Examples

Broadly applications are divided broadly into three types

i. Desktop Applications
ii. Mobile Applications
iii. Web Applications

Desktop Applications – Desktop Applications are applications that runs stand alone in a desktop or laptop computer.

Eg. Microsoft Paint

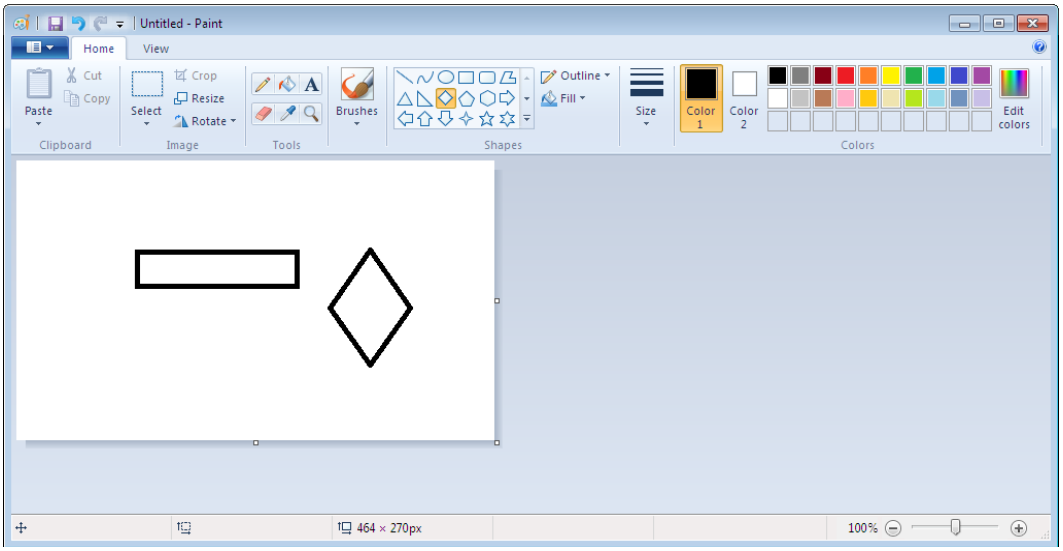


Figure 8

Figure 9

Mobile Applications

Desktop and laptop computers aren't the only devices that can run applications. You can also download applications for mobile devices like **smart phones** and **tablet computers**, which opens up a lot of new possibilities. Here are a few examples of mobile applications:

Google Wallet : is a mobile payment application

Figure 10

ICICI Mobile Banking Application

Figure 11

Web Applications - are applications that are accessed over a network such as the Internet or an Intranet.

Eg : ICICI Net Banking application

Figure 12

Passport Seva Portal to deliver passport services to citizens in a timely transparent

manner.

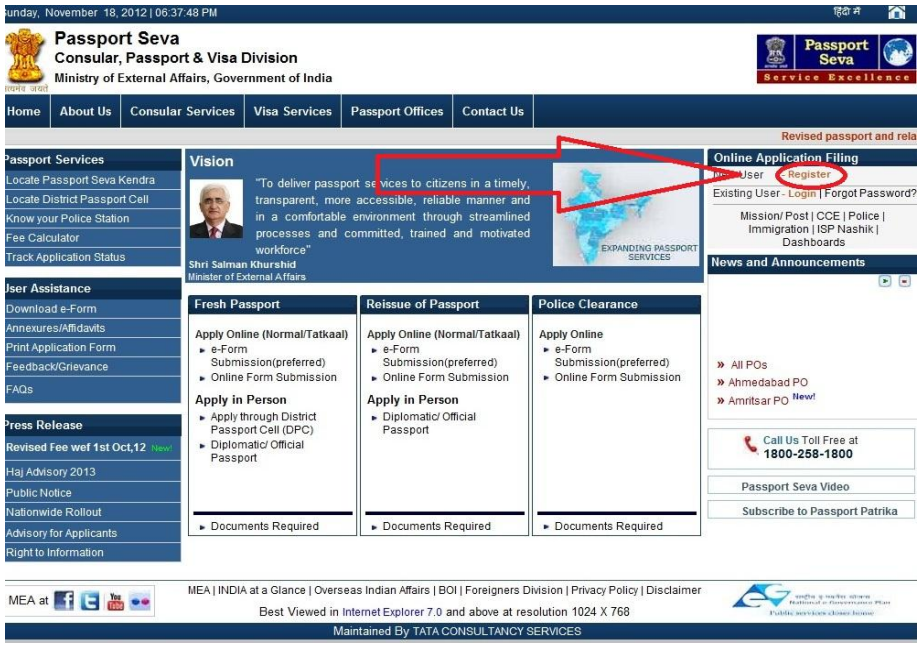


Figure 13

1.2.3 Basic Application Architecture

At the highest and most abstract level, the logical architecture view of any system can be considered as a set of cooperating components grouped into layers as given in the below figure.

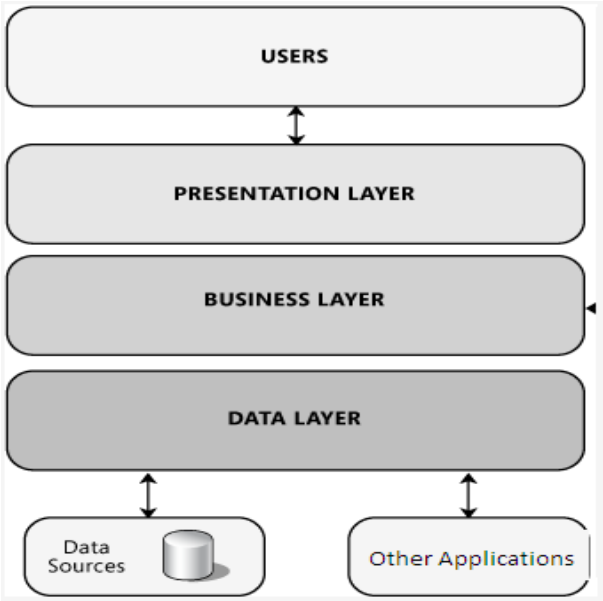


Figure 14

Presentation Layer / User Interface layer -

The top most layer of the application through which end-user interacts with the system. The main function of this layer is to translate the tasks and results to something which the end user can understand.

Business layer. This layer implements the core functionality of the system, and encapsulates the relevant business logic.

Data/Resource layer. This layer stores and retrieves information from the storage systems (databases/file systems). The information is then passed to the Business layer for processing and then eventually back to the user.

1.2.4 Layers and Tiers of an Application

Layer

Irrespective of the type of application that you are designing, and whether it has a user interface or it is a services application that only exposes services (not to be confused with services layer of an application), you can decompose the design into logical groupings of software components. These logical groupings are called layers. Layers help to differentiate between the different kinds of tasks performed by the components, making it easier to create a design that supports re-usability of components.

Benefits of layers:

Simplicity: easy to design once layers and their interaction are defined clearly
Flexibility: easy to modify and develop networks by separate layers modifications
Incremental changes: add new layers, add new functions to a layer

Tier – When the logical groups of components are physically separated in different machines, it is known as Tier. The communication between the different tiers of an application is through network.

Single/One Tier Architecture – All layers (Presentation, Business and Resource) are at same place. The application doesnot require network connection to work. It is easy to install in a system and use it.

Figure 15

Eg: MS Access, MS Excel

2- Tier Architecture - In 2 tier architecture there can be 2 possibilities

1. Presentation Layer in one machine and Business & Resource Layer is available in different machine
2. Presentation & Business layer in one machine and Resource layer in another machine.

Below figure shows a big picture of 2 tier Architecture.

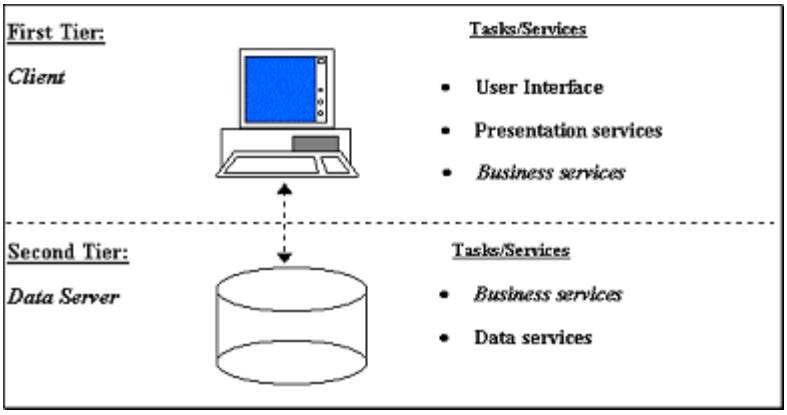


Figure 16

3 – Tier Architecture – In this all three layers of an application are physically separated in different machines.

Eg : Most of the web applications which involves data storage are 3 -tier architecture applications.

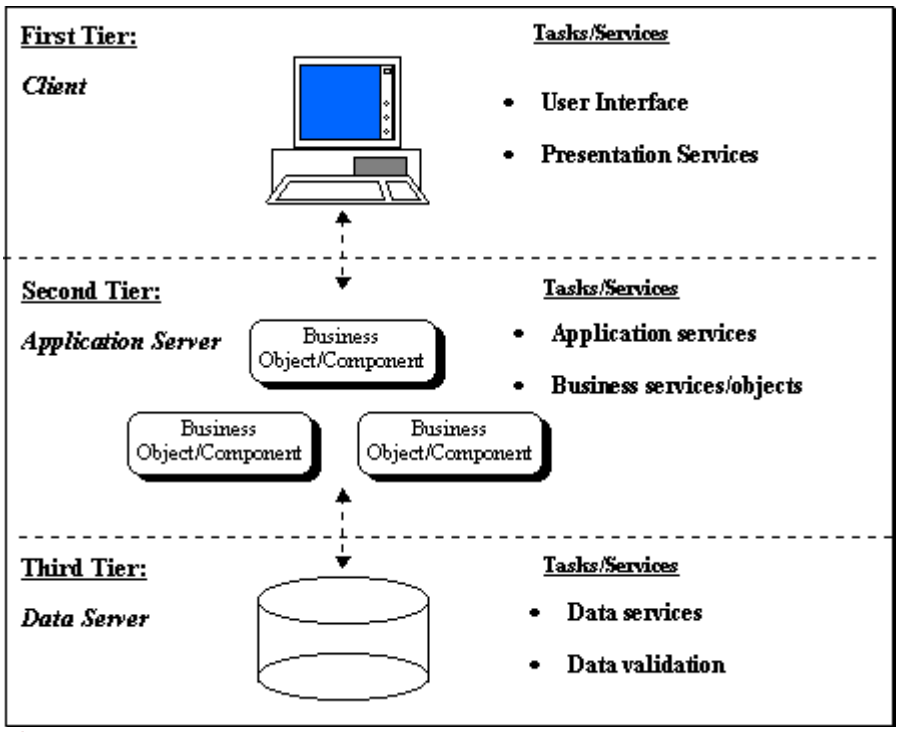


Figure 17

The below figure will help you in visualizing the communication in a web application.
In this example a 3-tier web application system is represented where all three layers of the application is distributed in three different machines.

The Presentation layer is at the client tier and from there request is sent to the Business layer and after processing the request its again sent to Storage/Resource tier for storing.

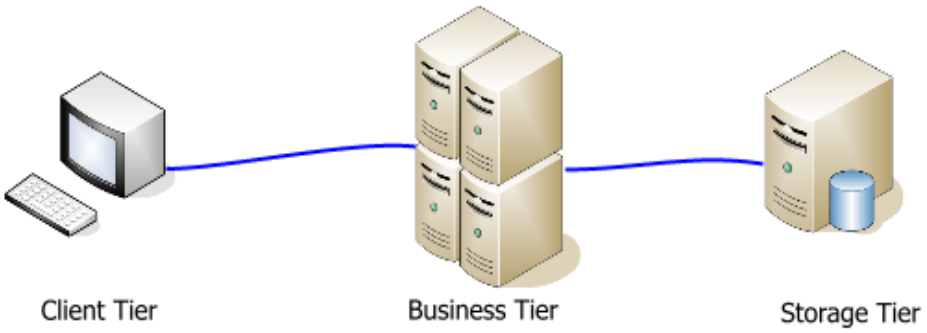


Figure 18

Multi-tier/ N-tier Architecture: All the 3 -tiered applications can be again physically separated by breaking it into different tiers and establishing a communication between all those tiers.

The terms tier and layer are often used interchangeably. When I use tier, I am referencing a physical layer. When I use layer, I am referencing a distinct software layer. Layers must exist on a tier, and thus tiers contain layers.

1.2.5 Basic Application Functions – CRUD

Each application gives the CRUD feature in its own way. CRUD is the acronym for Create, Read, Update and Delete.

Create – Adding/Storing any kind of data to the application

Read - There should be a feature where by the data added can be read

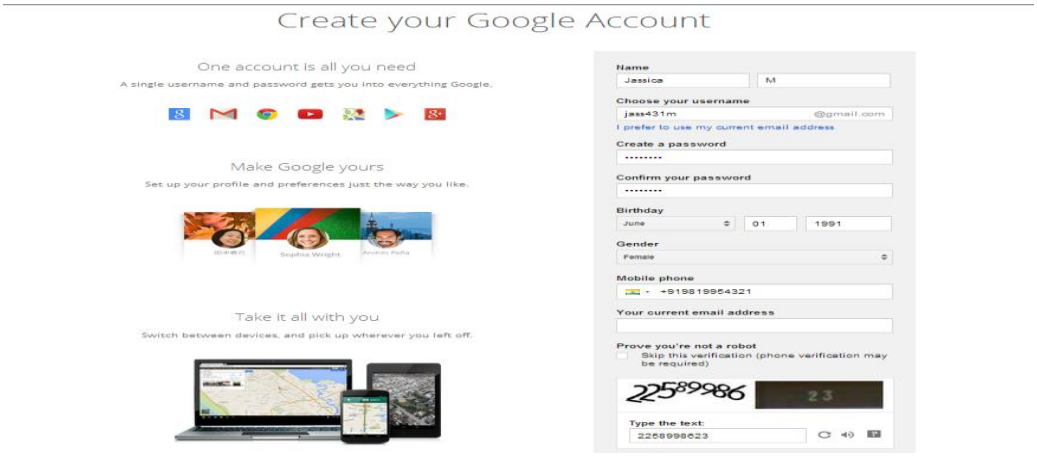
Update – There should be a facility to update the data created. Which user can perform this operation will depend purely on the application.

Delete – There should be a way to remove the data created. Again like Update, who can perform delete operation will vary depending on the application.

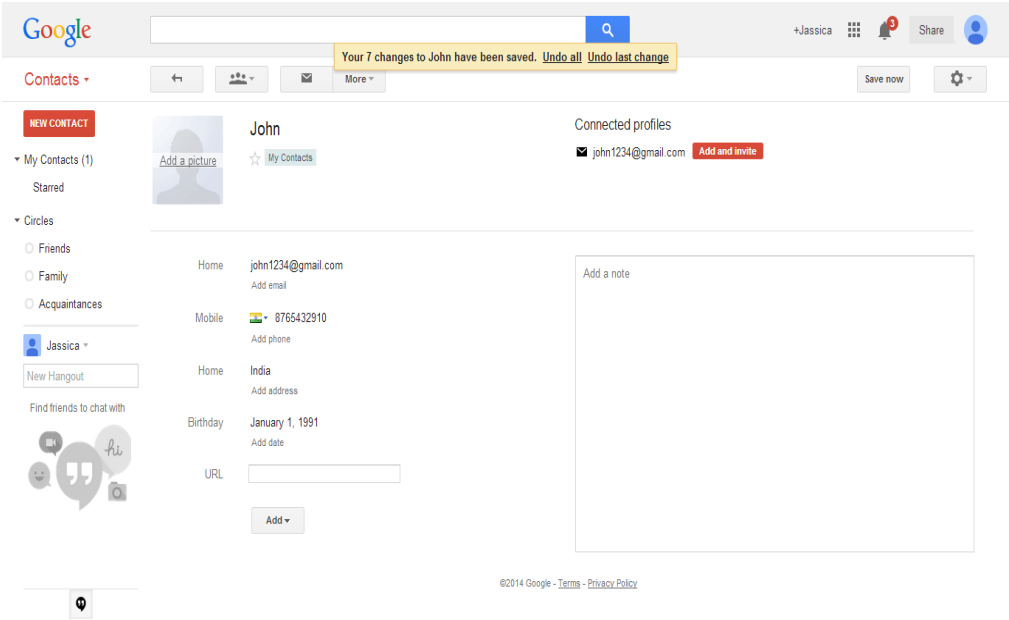
Remember that Application should follow CRUD, but not CURD ie, Create will happen first, Update and Delete should happen only after the data is Read.

1.2.5.1 Create

Let's take example of well-known website Google. When you first time register or create an account in this website, you have to fill certain details and click on register or sign up button. On clicking on the button, information entered by you will be verified and added into the database and your account will be created. This is one of the example for CREATE operation.



Similarly, adding a contact in gmail application of your Google account is also an example of create operation.



You can find these kind of options in many other websites. Creating your Facebook, twitter, IRCTC account, registering in naukary.com, even creating an event in your Facebook account are all examples of create operation.

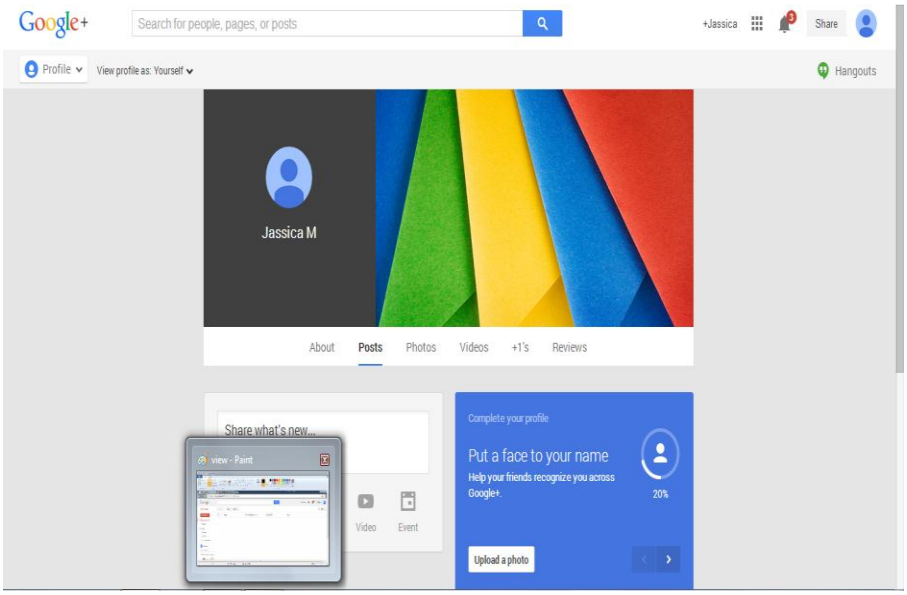
Whenever a new set of information is added into the database of the system, then it is a create operation.

It results in creation of new entry into the database of that web application.

1.2.5.2 Read

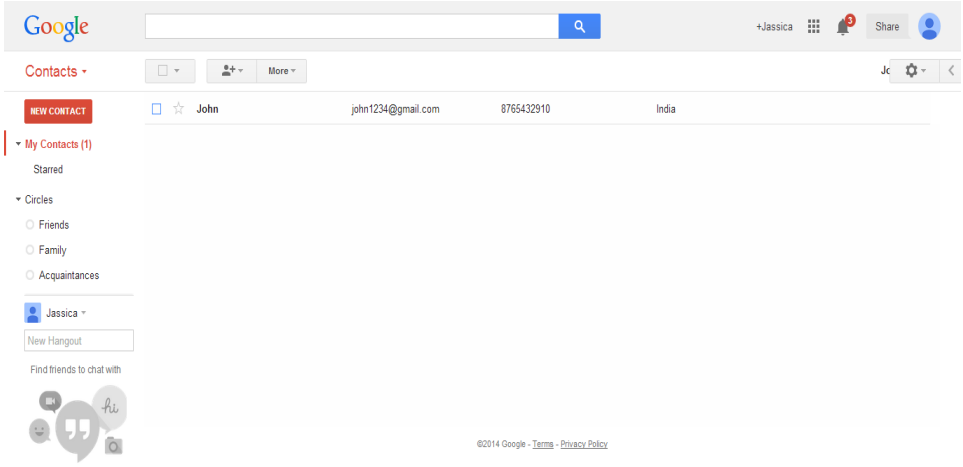
Let's take the example of Google again. When you log in to your Google account, a home page appears. It is a read operation. In this, the web application retrieves the information from the application's database, corresponding to the log in credentials which you have provided and displays it.

Viewing your profile by clicking on view profile is also an example of read operation.



When you are searching a person by their name, it is also a read operation. In this the web application will search for various user profiles with that particular name and displays the list.

Even viewing your contact list of gmail is an example of read.

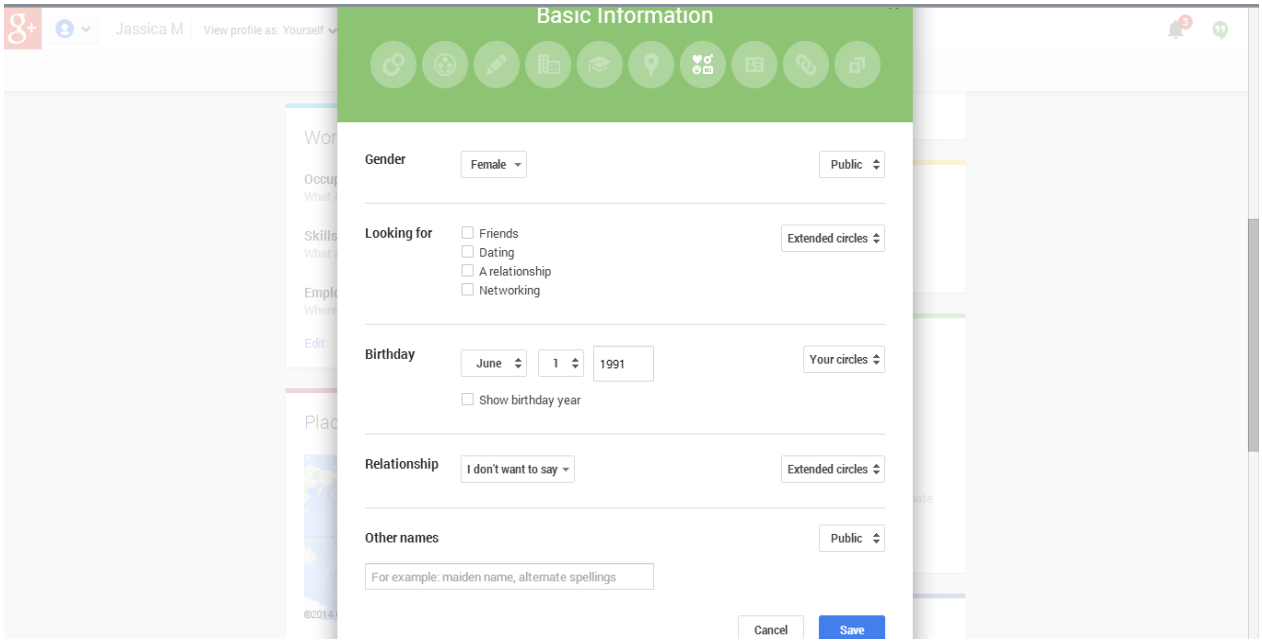


You can relate these examples with options being provided by many other social networks and web applications.

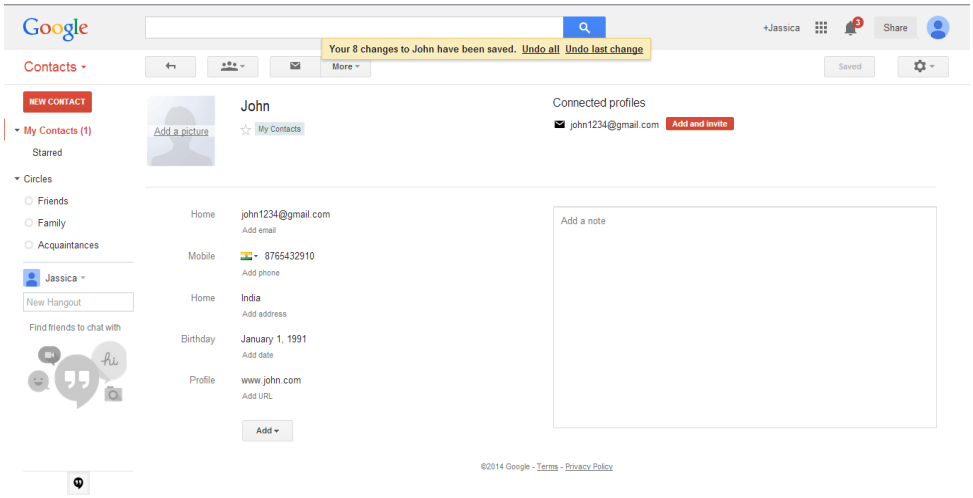
It shows that, read is basically, retrieving data from the database and displaying it to the user.

1.2.5.3 Update

Think, you want to change some of your basic information in your Google account, then, you will click on the edit button, which will take you to the edit profile page. It will display all the information which you have already added. You can enter the new information and by clicking on save, the information will get updated. Old information gets overwritten by the new information in the database.



Editing the contact details in gmail account is another example of update operation.

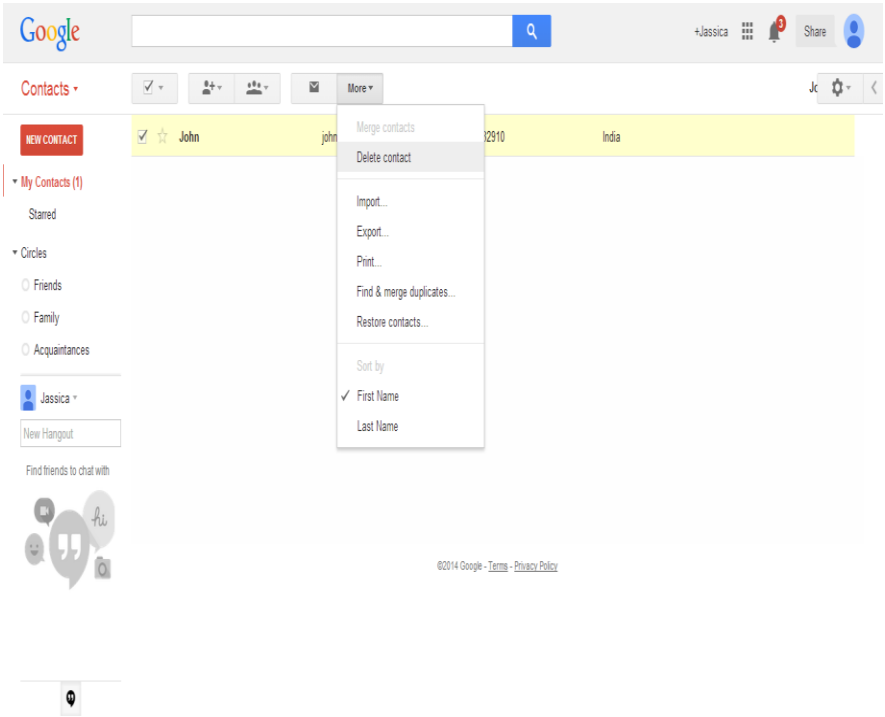


First time while filling the information, you might have left few non-mandatory fields blank. Filling those information later is also an example of update operation. Even removing few information (not all) like, phone number from your profile is also an example of update. Here the particular information will be set to blank.

Basically, update is, first a read operation, which displays the information which is already present in the database of the web application and then overwrites it with a new information.

1.2.5.4 Delete

Let us take the example of contacts in gmail. If you want to delete a contact from the contacts list, you can select the particular contact and click on the delete button. It will delete the contact from the contacts list.



If in future, you do not wish to use the account which you have created, then you can go for deactivation of the account. This is also an example of the delete operation.
Now a day most of the websites do not provide direct deletion, they allow you to just deactivate the account. Few websites even have facility to delete the account.

Delete is basically, deleting the information from the database of the website or making the information inaccessible.